SADI
Semantic Web Services

London, UK

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SADI Semantic Web Services

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http:// sadiframework.org/training/

canarie

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2.1 Introduction to Semantic Web Services

2.1 Introduction to Semantic Web Services

2.1.1 Standards for web services

2.1.2 Bioinformatics web services

2.1.3 Motivation for SADI

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2.1.1 Standards for Web Services

- WSDL
- SOAP
- UDDI
- SAWSDL
- OWL-S

2.1.2 Bioinformatics Web Services

- **tend to be simpler than the general web service case**
  - mostly stateless, atomic
    (excepting asynchronous operation)
  - data-centric
    retrieval
    analysis

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2.1.2 Bioinformatics Web Services

- **tend to ignore existing standards for semantic web services**
  - OWL-S
  - SAWSDL

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2.1.2 Bioinformatics Web Services

- **tend lately to reject even basic standards for web services**
  - SOAP

    (in favour of “RESTful services”)
  - UDDI

    (in favour of BioMoby, BioCatalogue, etc.)

2.1.3 Motivation for SADI

existing standards are too complicated

too much work for not enough gain

2.1.3 Motivation for SADI

- **simple**

  "perfection is finally attained not when there is no longer anything to add, but when there is no longer anything to take away"

- **powerful**

  "everything should be as simple as it can be, but not simpler"

- **flexible**

  "SADI: there's nothing to it"

2.2 Introduction to SADI Services

2.2 Introduction to SADI Services

2.2.1 What is a SADI service?

2.2.2 Why use SADI services?

2.2.3 Requirements for SADI compliance

2.2.4 Calling a SADI service

2.2.5 Advanced SADI features

2.2.6 The SADI registry

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2.2.1 What is a SADI service?

- **SADI services**
  - use simple HTTP operations
    (no project-specific protocol)
  - consume and produce RDF data
    (no project-specific message scaffolding)
  - use OWL to describe service interface
    (use existing ontologies
    and especially existing properties)

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2.2.1 What is a SADI service?

- **input is an RDF graph**
  - rooted at a particular node
  - described by an OWL class
- **output is an RDF graph**
  - rooted at the same node
  - described by another OWL class

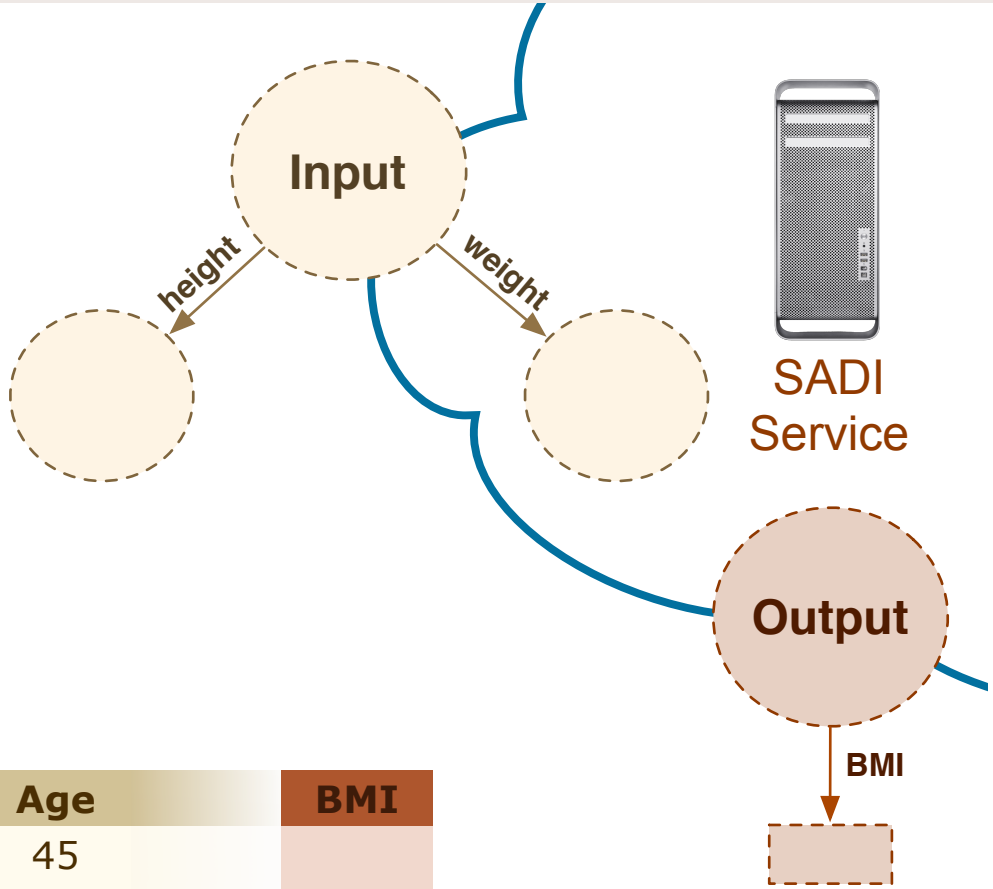
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2.2.1 What is a SADI service?

- **explicit relationship between input and output**
  - transparent service operation semantics
- **machine-understandable description of the service interface**
  - easy service discovery and integration

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2.2.2 Why use SADI services?

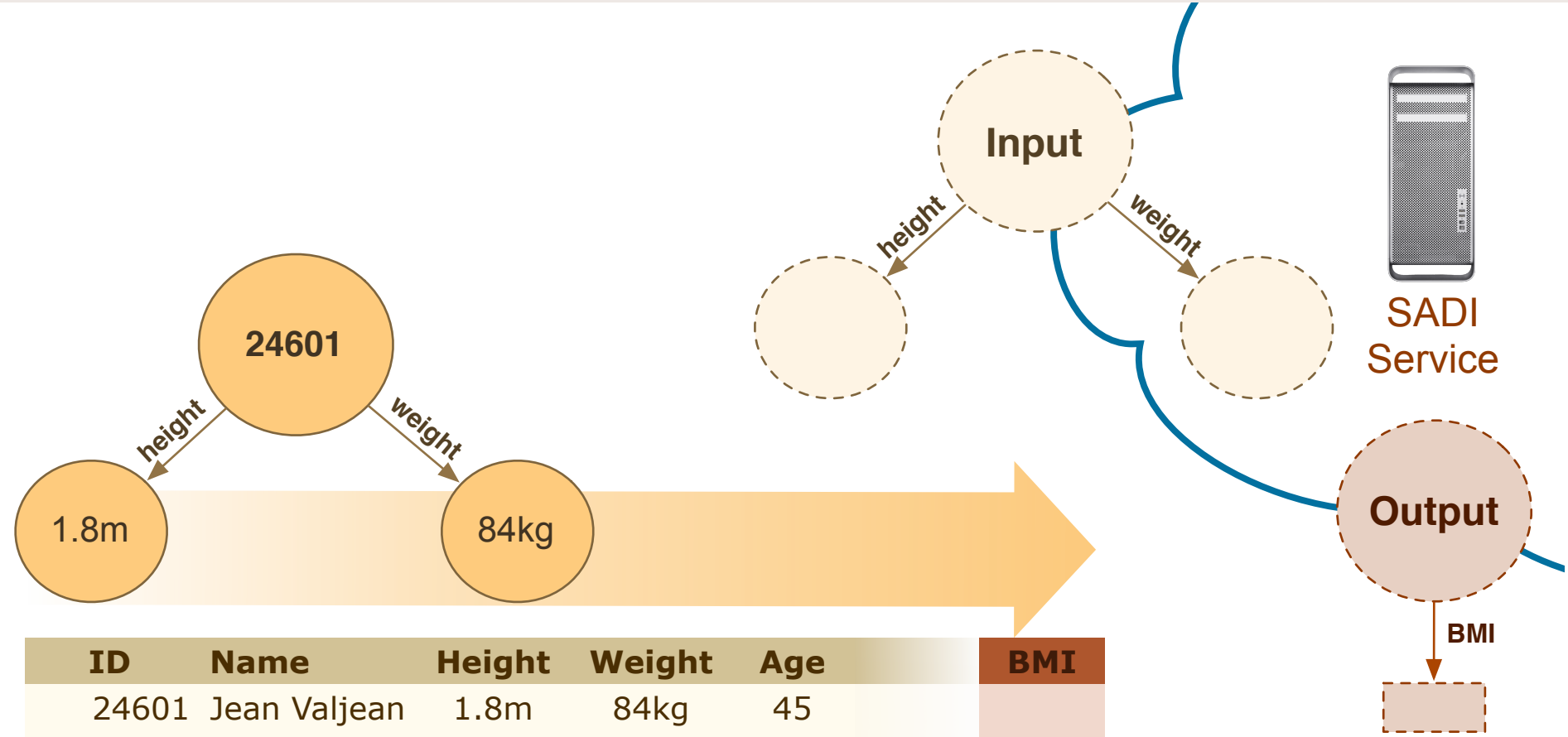


ID	Name	Height	Weight	Age	BMI
24601	Jean Valjean	1.8m	84kg	45	
7474505B	Jake Blues	1.73m	101kg	31	
6	—	1.88m	75kg	39	
	...	...	...	...	


Client

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2.2.2 Why use SADI services?

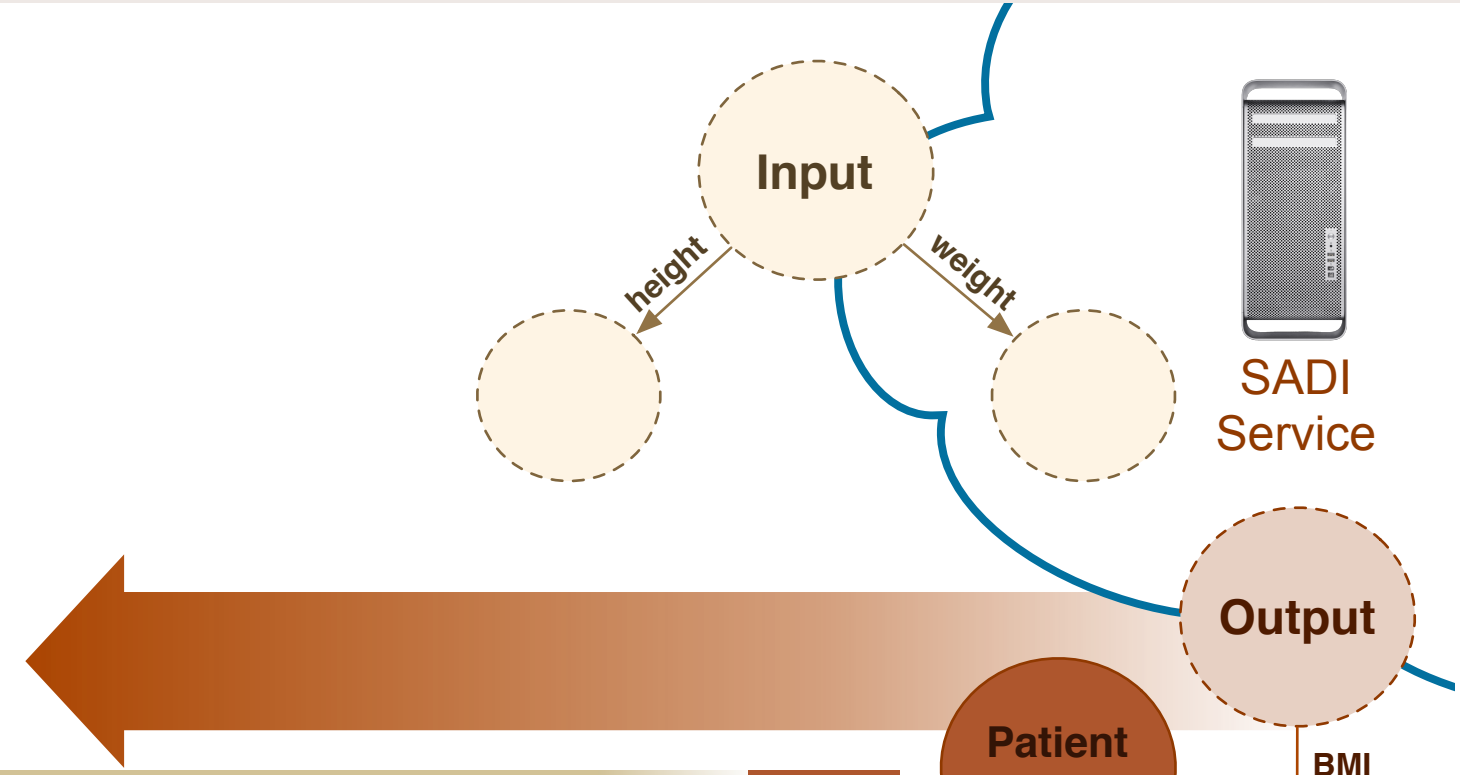


ID	Name	Height	Weight	Age	BMI
24601	Jean Valjean	1.8m	84kg	45	
7474505B	Jake Blues	1.73m	101kg	31	
6	—	1.88m	75kg	39	
	...	...	...	...	

Client

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2.2.2 Why use SADI services?



ID	Name	Height	Weight	Age	BMI
24601	Jean Valjean	1.8m	84kg	45	25.9
7474505B	Jake Blues	1.73m	101kg	31	
6	—	1.88m	75kg	39	
	...	...	...	...	


Client

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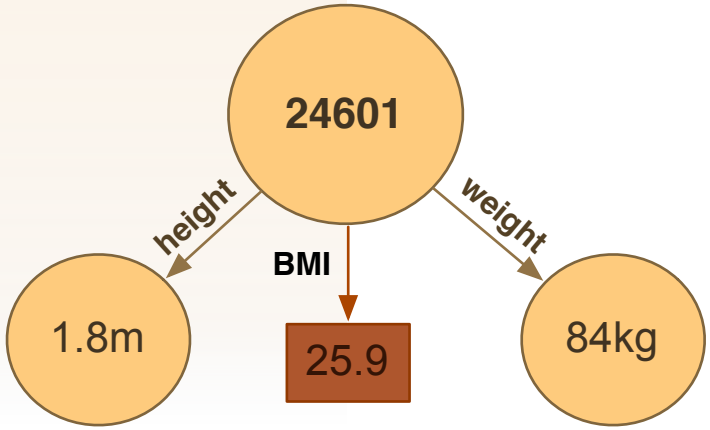
2.2.2 Why use SADI services?

```
@prefix rdf:     <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs:    <http://www.w3.org/2000/01/rdf-schema#> .
@prefix xsd:     <http://www.w3.org/2001/XMLSchema#> .
@prefix mged:    <http://mged.sourceforge.net/ontologies/MGEDOntology.owl#> .
@prefix foaf:    <http://xmlns.com/foaf/0.1/> .

<http://sadiframework.org/examples/bmi-input.rdf#patient24601>
      bmi:BMI "25.9"^^xsd:double ;
      foaf:name "Jean Valjean"^^xsd:string ;
      mged:has_height
              [ a       mged:Measurement ;
                mged:has_units mged:m ;
                mged:has_value "1.8"^^xsd:double
              ] ;
      mged:has_mass
              [ a       mged:Measurement ;
                mged:has_units mged:kg ;
                mged:has_value "84"^^xsd:double
              ] .
```



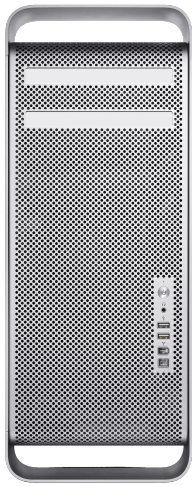
Client



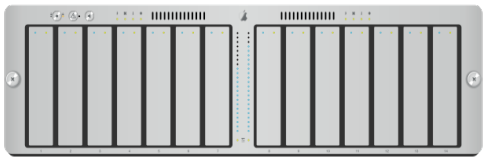
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2.2.2 Why use SADI services?

```
PREFIX sadi: <http://sadiframework.org/ontologies/predicates.owl#>
PREFIX dumontier: <http://ontology.dumontierlab.com/>
PREFIX kegg: <http://lsrn.org/KEGG_PATHWAY:>
SELECT ?gene ?protein ?seq
WHERE {
    kegg:hsa00232 dumontier:hasParticipant ?gene .
    ?gene sadi:encodes ?protein .
    ?protein sadi:hasSequence ?seq
}
```



Service



Service



Service

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2.2.3 Requirements for SADI compliance

- **Identified by an HTTP URL**
  - used to identify the service (service URI)
  - used to invoke the service (service URL)

http://sadiframework.org/examples/hello

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2.2.3 Requirements for SADI compliance

- **Responds to HTTP GET with service interface definition**
  - RDF document
  - service definition instance identified by service URI

    (see §2.3.1 for specifics)
  - indicates input OWL class and output OWL class

    (among other things)

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2.2.3 Requirements for SADI compliance

http://sadiframework.org/examples/hello

input OWL class is

`http://sadiframework.org/examples/hello.owl#NamedIndividual`

output OWL class is

`http://sadiframework.org/examples/hello.owl#GreetedIndividual`

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2.2.3 Requirements for SADI compliance

- **Responds to HTTP POST by invoking service**
  - input is an RDF document
    contains input OWL class instances
  - output is an RDF document
    contains output OWL class instances
    one for each input instance
  - input/output instance share same URI

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2.2.3 Requirements for SADI compliance

http://sadiframework.org/examples/hello

input:

```
@prefix hello: <http://sadiframework.org/examples/hello.owl#> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:NamedIndividual ;
   foaf:name "Guy Incognito" .
```

output:

```
@prefix hello: <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:GreetedIndividual ;
   hello:greeting "Hello, Guy Incognito!" .
```

2.2.3 Requirements for SADI compliance

- **Indicates an error with appropriate HTTP status**
  - usually 500 Internal Service Error
  - may return RDF describing the error

2.2.4 Calling a SADI service

1. **Client sends POST request to service URL**
   - POST body is an RDF document
     contains **input OWL class** instances
   - input serialization specified in `Content-type` HTTP header
     `application/rdf+xml`
     `text/rdf+n3`
   - output serialization requested in `Accept` HTTP header

2.2.4 Calling a SADI service

```
POST /examples/hello HTTP/1.1
Host: sadiframework.org
Content-type: text/rdf+n3
Accept: text/rdf+n3

@prefix hello: <http://sadiframework.org/examples/hello.owl#> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:NamedIndividual ;
   foaf:name "Guy Incognito" .
```

2.2.4 Calling a SADI service

2. Service responds with output

- response body is an RDF document
  - contains **output OWL class** instances
- serialization specified in `Content-type` HTTP header
  - `application/rdf+xml`
  - `text/rdf+n3`

2.2.4 Calling a SADI service

```
HTTP/1.1 200 OK
Content-type: text/rdf+n3

@prefix hello: <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:GreetedIndividual ;
   hello:greeting "Hello, Guy Incognito!" .
```

2.2.5 Advanced SADI features

2.2.5 Advanced SADI features

2.2.5.a Asynchronous SADI services

2.2.5.b Parameterized SADI services

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2.2.5.a Asynchronous SADI services

- **Problem**
  - HTTP time out if a service takes too long to respond
- **Solution**
  - service returns stub immediately; client polls for complete response

2.2.5.a Asynchronous SADI services

1. **Client sends POST request to service URL**
   - POST body is an RDF document
     contains input OWL class instances
   - input serialization specified in `Content-type` HTTP header
     `application/rdf+xml`
     `text/rdf+n3`
   - output serialization requested in `Accept` HTTP header

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2.2.5.a Asynchronous SADI services

```
POST /examples/hello HTTP/1.1
Host: sadiframework.org
Content-type: text/rdf+n3
Accept: text/rdf+n3

@prefix hello: <http://sadiframework.org/examples/hello.owl#> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:NamedIndividual ;
   foaf:name "Guy Incognito" .
```

2.2.5.a Asynchronous SADI services

2. Service responds with stub

- response code is `202 Accepted but incomplete`
- response body is an RDF document
  - contains **output OWL class** instances
  - contains `rdfs:isDefinedBy` statements
- serialization specified in `Content-type` HTTP header
  - `application/rdf+xml`
  - `text/rdf+n3`

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2.2.5.a Asynchronous SADI services

```
HTTP/1.1 202 Accepted
Content-type: text/rdf+n3

@prefix hello: <http://sadiframework.org/examples/hello.owl#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

<http://sadiframework.org/examples/hello-input.rdf#1>
  a hello:GreetedIndividual ;
  rdfs:isDefinedBy <http://sadiframework.org/examples/hello?poll=1> .
```

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2.2.5.a Asynchronous SADI services

3. Client fetches asynchronous data

- fetch the object URL of each `rdfs:isDefinedBy` statement

```
GET /examples/hello?poll=1 HTTP/1.1
Host: sadiframework.org
```

2.2.5.a Asynchronous SADI services

4. Until output is ready

- service sends an HTTP 302 redirect
  - Retry-after HTTP header suggests how long the client should wait (in seconds)

```
HTTP/1.1 302 Moved Temporarily
Retry-After: 30
Location: http://sadiframework.org/examples/hello?poll=1
```

2.2.5.a Asynchronous SADI services

5. Client follows redirect

- after waiting as suggested by `Retry-after` HTTP header

```
GET /examples/hello?poll=1 HTTP/1.1
Host: sadiframework.org
```

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2.2.5.a Asynchronous SADI services

6. When output is ready

- response body is an RDF document

  should be combined with original response

```
HTTP/1.1 200 OK
Content-type: text/rdf+n3

@prefix hello: <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:GreetedIndividual ;
   hello:greeting "Hello, Guy Incognito!" .
```

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2.2.5.b Parameterized SADI Services

- **Problem**
  - service has secondary parameters that are not part of the input data
    e.g. BLAST service: substitution matrix, number of results to return, etc.
- **Solution**
  - service specifies a **parameter OWL class**
  - client sends an instance of that class along with the input instances

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2.2.5.b Parameterized SADI Services

http://sadiframework.org/examples/hello-param

input:

```
@prefix hello: <http://sadiframework.org/examples/hello.owl#> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:NamedIndividual ;
   foaf:name "Guy Incognito" .
[] a hello:SecondaryParameters ;
   hello:lang "fr" .
```

output:

```
@prefix hello: <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:GreetedIndividual ;
   hello:greeting "Bonjour, Guy Incognito!" .
```

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2.2.5.b Parameterized SADI Services

http://sadiframework.org/examples/hello-param

input:

```
@prefix hello: <http://sadiframework.org/examples/hello.owl#> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:NamedIndividual ;
   foaf:name "Guy Incognito" .
[] a hello:SecondaryParameters ;
   hello:lang "es" .
```

output:

```
@prefix hello: <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello-input.rdf#1>
   a hello:GreetedIndividual ;
   hello:greeting "Hola, Guy Incognito!" .
```

2.2.6 The SADI registry

- **clients need to find SADI services**
  - ideally just search the web for instances of the SADI service description class

    current technology isn't quite there
- **use the SADI registry**
  - SPARQL endpoint, Java and REST APIs
  - anyone can register a service
  - services are indexed by the properties they attach

2.2.6 The SADI registry

SADI registry

http://sadiframework.org/registry/

SPARQL endpoint

http://sadiframework.org/registry/sparql

REST API

http://sadiframework.org/registry/api/

2.3 SADI Service Ontologies

2.3 SADI Service Ontologies

2.3.1 Service definition

2.3.2 Input and output OWL classes

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2.3.1 Service Definition

- name
- description
- contact email address
- authoritative status
- input OWL class
- output OWL class
- parameter OWL class
- test cases

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2.3.1 Service Definition

```
@prefix mygrid: <http://www.mygrid.org.uk/mygrid-moby-service#> .
@prefix hello:  <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello>
    a mygrid:serviceDescription ;
    mygrid:hasServiceNameText
        "ParamaterizedHelloWorld"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:hasServiceDescriptionText
        "A \"Hello, world!\" service where the output language is specified in a
parameter"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:providedBy
        [ a mygrid:organisation ;
            dc:creator mccarthy@elmonline.ca ;
            mygrid:authoritative
                "false"^^<http://www.w3.org/2001/XMLSchema#boolean>
        ] .
    mygrid:hasOperation
        [ a mygrid:operation ;
            mygrid:inputParameter
                [ a mygrid:secondaryParameter ;
                    mygrid:objectType hello:SecondaryParameters
                ] ;
            mygrid:outputParameter
                [ a mygrid:parameter ;
                    mygrid:objectType hello:GreetedIndividual
                ]
] .
```

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2.3.1 Service Definition

```
@prefix mygrid: <http://www.mygrid.org.uk/mygrid-moby-service#> .
@prefix hello:  <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello>
    a mygrid:serviceDescription ;
    mygrid:hasServiceNameText
        "ParamaterizedHelloWorld"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:hasServiceDescriptionText
        "A \"Hello, world!\" service where the output language is specified in a
parameter"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:providedBy
        [ a mygrid:organisation ;
            dc:creator mccarthy@elmonline.ca ;
            mygrid:authoritative
                "false"^^<http://www.w3.org/2001/XMLSchema#boolean>
        ] .
    mygrid:hasOperation
        [ a mygrid:operation ;
            mygrid:inputParameter
                [ a mygrid:secondaryParameter ;
                    mygrid:objectType hello:SecondaryParameters
                ] ;
            mygrid:outputParameter
                [ a mygrid:parameter ;
                    mygrid:objectType hello:GreetedIndividual
                ]
] .
```

Name

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2.3.1 Service Definition

```
@prefix mygrid: <http://www.mygrid.org.uk/mygrid-moby-service#> .
@prefix hello:  <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello>
    a mygrid:serviceDescription ;
    mygrid:hasServiceNameText
        "ParamaterizedHelloWorld"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:hasServiceDescriptionText
        "A \"Hello, world!\" service where the output language is specified in a
parameter"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:providedBy
        [ a mygrid:organisation ;
            dc:creator mccarthy@elmonline.ca ;
            mygrid:authoritative
                "false"^^<http://www.w3.org/2001/XMLSchema#boolean>
        ] .
    mygrid:hasOperation
        [ a mygrid:operation ;
            mygrid:inputParameter
                [ a mygrid:secondaryParameter ;
                    mygrid:objectType hello:SecondaryParameters
                ] ;
            mygrid:outputParameter
                [ a mygrid:parameter ;
                    mygrid:objectType hello:GreetedIndividual
                ]
] .
```

Description

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2.3.1 Service Definition

```
@prefix mygrid: <http://www.mygrid.org.uk/mygrid-moby-service#> .
@prefix hello:  <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello>
    a mygrid:serviceDescription ;
    mygrid:hasServiceNameText
        "ParamaterizedHelloWorld"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:hasServiceDescriptionText
        "A \"Hello, world!\" service where the output language is specified in a
parameter"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:providedBy
        [ a mygrid:organisation ;
            dc:creator mccarthy@elmonline.ca ;
            mygrid:authoritative
                "false"^^<http://www.w3.org/2001/XMLSchema#boolean>
        ] .
    mygrid:hasOperation
        [ a mygrid:operation ;
            mygrid:inputParameter
                [ a mygrid:secondaryParameter ;
                    mygrid:objectType hello:SecondaryParameters
                ] ;
            mygrid:outputParameter
                [ a mygrid:parameter ;
                    mygrid:objectType hello:GreetedIndividual
                ]
] .
```

Contact email address

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2.3.1 Service Definition

```
@prefix mygrid: <http://www.mygrid.org.uk/mygrid-moby-service#> .
@prefix hello:  <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello>
    a mygrid:serviceDescription ;
    mygrid:hasServiceNameText
        "ParamaterizedHelloWorld"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:hasServiceDescriptionText
        "A \"Hello, world!\" service where the output language is specified in a
parameter"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:providedBy
        [ a mygrid:organisation ;
            dc:creator mccarthy@elmonline.ca ;
            mygrid:authoritative
                "false"^^<http://www.w3.org/2001/XMLSchema#boolean>
        ] .
    mygrid:hasOperation
        [ a mygrid:operation ;
            mygrid:inputParameter
                [ a mygrid:secondaryParameter ;
                    mygrid:objectType hello:SecondaryParameters
                ] ;
            mygrid:outputParameter
                [ a mygrid:parameter ;
                    mygrid:objectType hello:GreetedIndividual
                ]
] .
```

Authoritative

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2.3.1 Service Definition

```
@prefix mygrid: <http://www.mygrid.org.uk/mygrid-moby-service#> .
@prefix hello:  <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello>
    a mygrid:serviceDescription ;
    mygrid:hasServiceNameText
        "ParamaterizedHelloWorld"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:hasServiceDescriptionText
        "A \"Hello, world!\" service where the output language is specified in a
parameter"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:providedBy
        [ a mygrid:organisation ;
            dc:creator mccarthy@elmonline.ca ;
            mygrid:authoritative
                "false"^^<http://www.w3.org/2001/XMLSchema#boolean>
        ] .
    mygrid:hasOperation
        [ a mygrid:operation ;
            mygrid:inputParameter
                [ a mygrid:secondaryParameter ;
                    mygrid:objectType hello:SecondaryParameters
                ] ;
            mygrid:outputParameter
                [ a mygrid:parameter ;
                    mygrid:objectType hello:GreetedIndividual
                ]
] .
```

Input OWL class

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2.3.1 Service Definition

```
@prefix mygrid: <http://www.mygrid.org.uk/mygrid-moby-service#> .
@prefix hello:  <http://sadiframework.org/examples/hello.owl#> .

<http://sadiframework.org/examples/hello>
    a mygrid:serviceDescription ;
    mygrid:hasServiceNameText
        "ParamaterizedHelloWorld"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:hasServiceDescriptionText
        "A \"Hello, world!\" service where the output language is specified in a
parameter"^^<http://www.w3.org/2001/XMLSchema#string> ;
    mygrid:providedBy
        [ a mygrid:organisation ;
            dc:creator mccarthy@elmonline.ca ;
            mygrid:authoritative
                "false"^^<http://www.w3.org/2001/XMLSchema#boolean>
        ] .
    mygrid:hasOperation
        [ a mygrid:operation ;
            mygrid:inputParameter
                [ a mygrid:secondaryParameter ;
                    mygrid:objectType hello:SecondaryParameters
                ] ;
            mygrid:outputParameter
                [ a mygrid:parameter ;
                    mygrid:objectType hello:GreetedIndividual
                ]
] .
```

Output OWL class

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2.3.1 Service Definition

```
<http://sadiframework.org/examples/hello-param>
    ...
    mygrid:hasOperation
        [ a mygrid:operation ;
          ...
          mygrid:inputParameter
            [ a mygrid:secondaryParameter ;
              mygrid:objectType hello:ParameterClass
              mygrid:hasDefaultValue
                [ a hello:ParameterClass ;
                  hello:lang “en”
                ]
            ] .
          mygrid:hasUnitTest
            [ a mygrid:testCase ;
              mygrid:exampleInput
                [ a hello:InputClass ;
                  foaf:name “Guy Incognito”
                ] .
              mygrid:exampleOutput
                [ a hello:OutputClass ;
                  hello:greeting “Hello, Guy Incognito!”
                ] .
            ] .
          mygrid:hasUnitTest
            [ a mygrid:testCase ;
              mygrid:exampleInput test:hello-param-input.rdf ;
              mygrid:exampleOutput test:hello-param-output.rdf
            ]
        ]
```

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2.3.1 Service Definition

```
<http://sadiframework.org/examples/hello-param>
    ...
    mygrid:hasOperation
        [ a mygrid:operation ;
          ...
          mygrid:inputParameter
            [ a mygrid:secondaryParameter ;
              mygrid:objectType hello:ParameterClass
              mygrid:hasDefaultValue
                [ a hello:ParameterClass ;
                  hello:lang “en”
                ]
            ] .
          mygrid:hasUnitTest
            [ a mygrid:testCase ;
              mygrid:exampleInput
                [ a hello:InputClass ;
                  foaf:name “Guy Incognito”
                ] .
              mygrid:exampleOutput
                [ a hello:OutputClass ;
                  hello:greeting “Hello, Guy Incognito!”
                ] .
            ] .
          mygrid:hasUnitTest
            [ a mygrid:testCase ;
              mygrid:exampleInput test:hello-param-input.rdf ;
              mygrid:exampleOutput test:hello-param-output.rdf
            ]
        ]
```

Parameter OWL class

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2.3.1 Service Definition

```
<http://sadiframework.org/examples/hello-param>
    ...
    mygrid:hasOperation
        [ a mygrid:operation ;
          ...
          mygrid:inputParameter
            [ a mygrid:secondaryParameter ;
              mygrid:objectType hello:ParameterClass
              mygrid:hasDefaultValue
                [ a hello:ParameterClass ;
                  hello:lang “en”
                ]
            ] .
          mygrid:hasUnitTest
            [ a mygrid:testCase ;
              mygrid:exampleInput
                [ a hello:InputClass ;
                  foaf:name “Guy Incognito”
                ] .
              mygrid:exampleOutput
                [ a hello:OutputClass ;
                  hello:greeting “Hello, Guy Incognito!”
                ] .
            ] .
          mygrid:hasUnitTest
            [ a mygrid:testCase ;
              mygrid:exampleInput test:hello-param-input.rdf ;
              mygrid:exampleOutput test:hello-param-output.rdf
            ]
        ]
```

Test case

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2.3.1 Service Definition

```
<http://sadiframework.org/examples/hello-param>
    ...
    mygrid:hasOperation
        [ a mygrid:operation ;
          ...
          mygrid:inputParameter
            [ a mygrid:secondaryParameter ;
              mygrid:objectType hello:ParameterClass
              mygrid:hasDefaultValue
                [ a hello:ParameterClass ;
                  hello:lang “en”
                ]
            ] .
          mygrid:hasUnitTest
            [ a mygrid:testCase ;
              mygrid:exampleInput
                [ a hello:InputClass ;
                  foaf:name “Guy Incognito”
                ] .
              mygrid:exampleOutput
                [ a hello:OutputClass ;
                  hello:greeting “Hello, Guy Incognito!”
                ] .
            ] .
          mygrid:hasUnitTest
            [ a mygrid:testCase ;
              mygrid:exampleInput test:hello-param-input.rdf ;
              mygrid:exampleOutput test:hello-param-output.rdf
            ]
        ]
```

Test case

2.3.2 Input and output OWL classes

2.3.2 Input and output OWL classes

2.3.2.a Input OWL classes

2.3.2.b Output OWL classes

2.3.2.a Input OWL classes

- **property restrictions describe data consumed by the service**
- **all property restrictions must be satisfied by each input instance**
  - optional properties can be indicated by minimum cardinality of 0

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2.3.2.a Input OWL classes

- **instances should be dynamically identifiable**
  - use necessary and sufficient conditions
  - avoid universal and exact/maximum cardinality restrictions

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2.3.2.a Input OWL classes

http://sadiframework.org/examples/hello.owl#InputClass

Manchester OWL syntax:

```
Class: hello:NamedIndividual
    EquivalentTo:
        foaf:name min 1 rdfs:Literal
```

RDF/XML syntax:

necessary and sufficient

```
<owl:Class rdf:about="&hello;NamedIndividual">
  <owl:equivalentClass>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&foaf;name"/>
      <owl:minCardinality rdf:datatype="&xsd;integer">1</owl:minCardinality>
    </owl:Restriction>
  </owl:equivalentClass>
</owl:Class>
```

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2.3.2.a Input OWL classes

http://sadiframework.org/examples/hello.owl#BadInputClass

Manchester OWL syntax:

```
Class: hello:NamedIndividual
   EquivalentTo:
       foaf:name exactly 1 rdfs:Literal
```

RDF/XML syntax:

exact cardinality

```
<owl:Class rdf:about="&hello;NamedIndividual">
  <owl:equivalentClass>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&foaf;name"/>
      <owl:cardinality rdf:datatype="&xsd;integer">1</owl:cardinality>
    </owl:Restriction>
  </owl:equivalentClass>
</owl:Class>
```

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2.3.2.a Input OWL classes

http://sadiframework.org/examples/hello.owl#BadInputClass

Manchester OWL syntax:

```
Class: hello:NamedIndividual
    EquivalentTo:
        foaf:name exactly 1 rdfs:Literal
```

RDF/XML syntax:

exact cardinality

```
<owl:Class rdf:about="&hello;NamedIndividual">
  <owl:equivalentClass>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&foaf;name"/>
      <owl:cardinality rdf:datatype="&xsd;integer">1</owl:cardinality>
    </owl:Restriction>
  </owl:equivalentClass>
</owl:Class>
```

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2.3.2.a Input OWL classes

http://purl.org/SADI/LSRN/UniProt_Record

Manchester OWL syntax:

```
Class: lsrn:UniProt_Record
    EquivalentTo:
        sio:SIO_000008 some lsrn:UniProt_Identifier
```

RDF/XML syntax:

existential

```
<owl:Class rdf:about="UniProt_Record">
  <owl:equivalentClass>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&sio;SIO_000008"/>
      <owl:someValuesFrom rdf:resource="&lsrn;UniProt_Identifier"/>
    </owl:Restriction>
  </owl:equivalentClass>
</owl:Class>
```

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2.3.2.a Input OWL classes

http://purl.org/SADI/LSRN/Bad_UniProt_Record

Manchester OWL syntax:

```
Class: lsrn:Bad_UniProt_Record
    EquivalentTo:
        sio:SIO_000008 only lsrn:UniProt_Identifier
```

RDF/XML syntax:

universal

```
<owl:Class rdf:about="UniProt_Record">
  <owl:equivalentClass>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&sio;SIO_000008"/>
      <owl:allValuesFrom rdf:resource="&lsrn;UniProt_Identifier"/>
    </owl:Restriction>
  </owl:equivalentClass>
</owl:Class>
```

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2.3.2.a Input OWL classes

http://purl.org/SADI/LSRN/Bad_UniProt_Record

Manchester OWL syntax:

```
Class: lsrn:Bad_UniProt_Record
    EquivalentTo:
        sio:SIO_000008 only lsrn:UniProt_Identifier
```

RDF/XML syntax:

```
<owl:Class rdf:about="UniProt_Record">
  <owl:equivalentClass>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&sio;SIO_000008"/>
      <owl:allValuesFrom rdf:resource="&lsrn;UniProt_Identifier"/>
    </owl:Restriction>
  </owl:equivalentClass>
</owl:Class>
```

universal

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2.3.2.a Input OWL classes

http://sadiframework.org/examples/regression.owl#DatedValue

Manchester OWL syntax:

```
Class: regress:DatedValue
    EquivalentTo:
        (dc:date some xsd:date) and
        (rdf:value some xsd:double)
```

RDF/XML syntax:

necessary and sufficient (multiple properties)

```
<owl:Class rdf:about="&regression;DatedValue">
  <owl:intersectionOf rdf:parseType="Collection">
    <owl:Restriction>
      <owl:onProperty rdf:resource="&dc;date"/>
      <owl:someValuesFrom rdf:resource="&xsd;date"/>
    </owl:Restriction>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&rdf;value"/>
      <owl:someValuesFrom rdf:resource="&xsd;double"/>
    </owl:Restriction>
  </owl:intersectionOf>
</owl:Class>
```

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2.3.2.a Input OWL classes

http://sadiframework.org/ontologies/samples.owl#InputClass

Manchester OWL syntax:

```
Class: sample:InputClass
    EquivalentTo:
        (sample:required min 1 owl:Thing) and
        (sample:optional min 0 owl:Thing)
```

RDF/XML syntax:

required property (min cardinality 1)

```
<owl:Class rdf:about="#InputClass">
    <owl:intersectionOf rdf:parseType="Collection">
      <owl:Restriction>
        <owl:onProperty rdf:resource="#required"/>
        <owl:minCardinality rdf:datatype="&xsd;integer">1</owl:minCardinality>
      </owl:Restriction>
      <owl:Restriction>
        <owl:onProperty rdf:resource="#optional"/>
        <owl:minCardinality rdf:datatype="&xsd;integer">0</owl:minCardinality>
      </owl:Restriction>
    </owl:intersectionOf>
  </owl:Class>
```

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2.3.2.a Input OWL classes

http://sadiframework.org/ontologies/samples.owl#InputClass

Manchester OWL syntax:

```
Class: sample:InputClass
    EquivalentTo:
        (sample:required min 1 owl:Thing) and
        (sample:optional min 0 owl:Thing)
```

RDF/XML syntax:

optional property (min cardinality 0)

```
<owl:Class rdf:about="#InputClass">
    <owl:intersectionOf rdf:parseType="Collection">
      <owl:Restriction>
        <owl:onProperty rdf:resource="#required"/>
        <owl:minCardinality rdf:datatype="&xsd;integer">1</owl:minCardinality>
      </owl:Restriction>
      <owl:Restriction>
        <owl:onProperty rdf:resource="#optional"/>
        <owl:minCardinality rdf:datatype="&xsd;integer">0</owl:minCardinality>
      </owl:Restriction>
    </owl:intersectionOf>
  </owl:Class>
```

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2.3.2.b Output OWL classes

- **property restrictions describe data produced by the service**
- **range of attached data should be indicated on the property restriction**
  - so clients can discriminate between services that attach the same properties

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2.3.2.b Output OWL classes

- **instances don't have to be dynamically identifiable**
  - no harm if they are
  - may be preferred for consistency with input OWL class

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2.3.2.b Output OWL classes

http://sadiframework.org/examples/keggPathway2Gene.owl#OutputClass

Manchester OWL syntax:

```
Class: OutputClass
    SubClassOf:
        sadi:hasParticipant some lsrn:KEGG
```

RDF/XML syntax:

necessary
but not sufficient

```
<owl:Class rdf:about="#OutputClass">
  <rdfs:subClassOf>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&sadi;hasParticipant"/>
      <owl:someValuesFrom rdf:resource="&lsrn;KEGG"/>
    </owl:Restriction>
  </rdfs:subClassOf>
</owl:Class>
```

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2.3.2.b Output OWL classes

http://sadiframework.org/examples/keggPathway2Gene.owl#OutputClass

Manchester OWL syntax:

```
Class: OutputClass
    SubClassOf:
        sadi:hasParticipant some lsrn:KEGG
```

RDF/XML syntax:

```
<owl:Class rdf:about="#OutputClass">
  <rdfs:subClassOf>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&sadi;hasParticipant"/>
      <owl:someValuesFrom rdf:resource="&lsrn;KEGG"/>
    </owl:Restriction>
  </rdfs:subClassOf>
</owl:Class>
```

ranged restriction (KEGG genes)

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2.3.2.b Output OWL classes

http://sadiframework.org/examples/keggPathway2Compound.owl#OutputClass

Manchester OWL syntax:

```
Class: OutputClass
    SubClassOf:
        sadi:hasParticipant some lsrn:KEGG_Compound
```

RDF/XML syntax:

```
<owl:Class rdf:about="#OutputClass">
  <rdfs:subClassOf>
    <owl:Restriction>
      <owl:onProperty rdf:resource="&sadi;hasParticipant"/>
      <owl:someValuesFrom rdf:resource="&lsrn;KEGG_Compound"/>
    </owl:Restriction>
  </rdfs:subClassOf>
</owl:Class>
```

ranged restriction (KEGG compounds)

2.4 SADI Service Generator

2.4.1 Exercise

2.4.1 Exercise

Create a SADI service that takes height and weight as input and attaches BMI.

2.4.1 Exercise

- **Create input and output OWL classes**
  - Protégé
- **Generate SADI service skeleton**
  - SADI Protégé plugin
  - generate Java or Perl code
- **Add service logic**
  - Eclipse + Maven plugin (for Java code)
  - your favourite editor (for Perl code)
- **Validate and register service**

2.5 SADI and SAWSDL

2.5 SADI and SAWSDL

2.5.1 Introduction to SAWSDL

2.5.2 Using SAWSDL with SADI

2.5.1 Introduction to SAWSDL

- **Semantic Annotations for WSDL**
  - annotations for input, output, operation elements of WSDL
  - sawsdl:modelReference attribute maps WSDL element to ontology element
  - sawsdl:loweringSchemaMapping maps RDF to XML (for input)
  - sawsdl:liftingSchemaMapping maps XML to RDF (for output)

2.5.1 Introduction to SAWSDL

- **lifting schema**
  - “lifts” non-semantic XML into semantic output
  - standard imposes no requirements
  - W3C suggests XSLT

2.5.1 Introduction to SAWSDL

- **lowering schema**
  - "lowers" semantic input into non-semantic XML
  - standard imposes no requirements
  - XSLT is insufficient
    (RDF > XML)
  - W3C suggests SPARQL + XSLT
    (styled SPARQL results XML)

2.5.1 Introduction to SAWSDL

RDF > XML

```
<hello:NamedIndividual rdf:about="#Guy">
    <foaf:name>Guy Incognito</foaf:name>
    <foaf:knows>
        <hello:NamedIndividual rdf:about="#JoeyJoJo"/>
        <foaf:name>Joey Jo-Jo Jr. Shabadoo</foaf:name>
    </foaf:knows>
</hello:NamedIndividual>
```

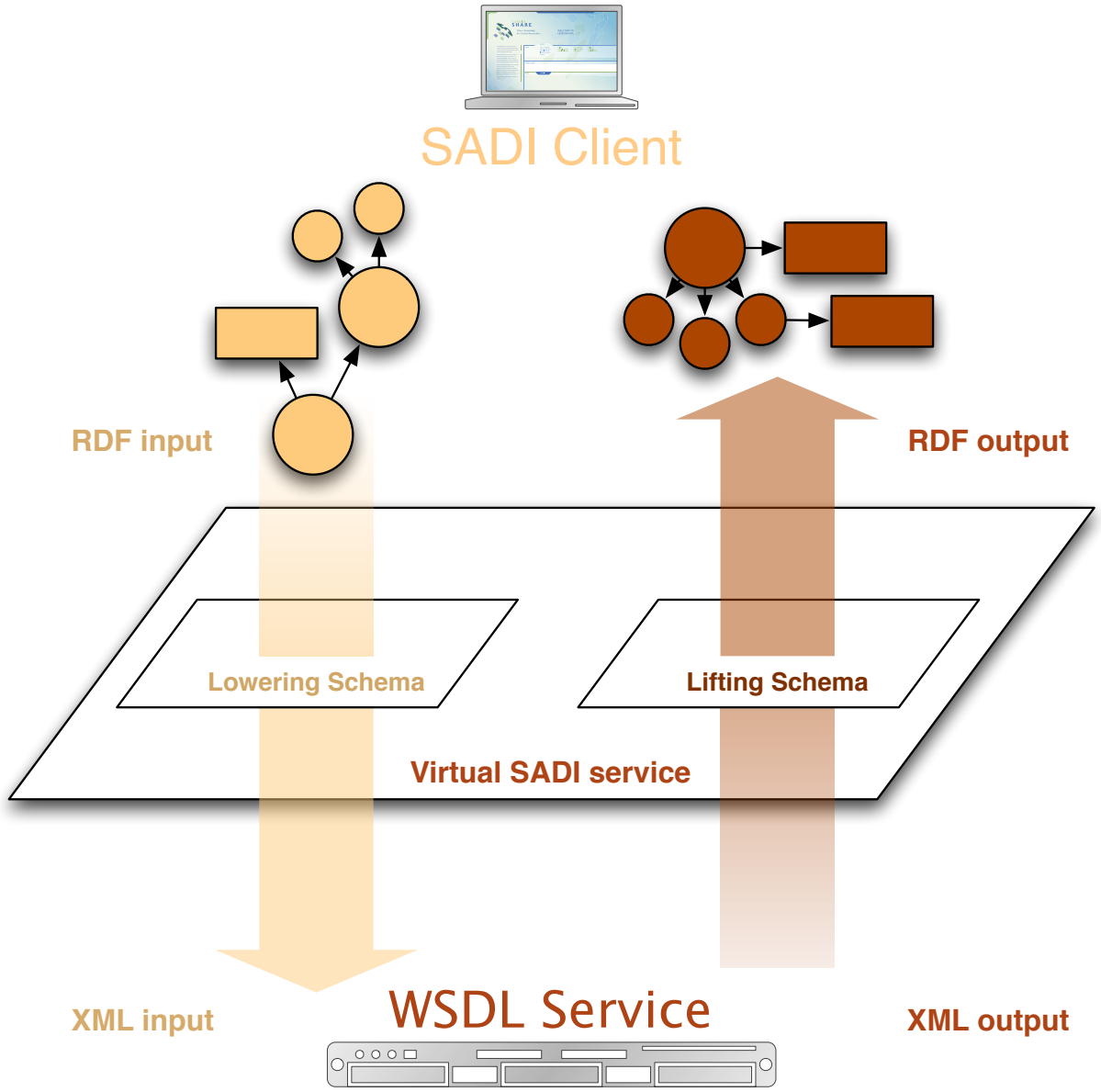
```
<rdf:Description rdf:about="#Guy">
    <rdf:type rdf:resource="#NamedIndividual"/>
    <foaf:name>Guy Incognito</foaf:name>
    <foaf:knows rdf:resource="#JoeyJoJo"/>
</rdf:Description>

<rdf:Description rdf:about="#JoeyJoJo">
    <rdf:type rdf:resource="#NamedIndividual"/>
    <foaf:name>Joey Jo-Jo Jr. Shabadoo</foaf:name>
</rdf:Description>
```

...

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2.5.2 Using SAWSDL with SADI



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2.5.2 Using SAWSDL with SADI

- **SADI lifting schema options**
  - XSLT
  - XPath + RDFPath
  - ...
- **SADI lowering schema options**
  - SPARQL + XSLT
  - SPARQL + Velocity
  - ...

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2.5.2 Using SAWSDL with SADI

- **SADI SAWSDL generator**
  - generate SAWSDL from existing WSDL service
  - automatically generate SADI interface, including input/output OWL classes and a SADI-compliant service endpoint

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2.5.2 Using SAWSDL with SADI

- **Manual SAWSDL generation**
  - modelReferences are SADI-compliant input and output OWL classes
  - lifting and lowering schemata are valid SADI options
  - register SAWSDL as a virtual SADI service

    (no SADI-compliant service endpoint;
    service is only accessible through client API)

2.6 SADI Clients

2.6 SADI Clients

2.6.1 SADI client API

2.6.2 SHARE query client

2.6.3 SADI Taverna plugin

2.6.1 SADI client API

- **Java library**
  (available via Maven)
- **Find services**
  - by attached properties
  - by input data
- **Invoke services**
- **Semantic utility functions**

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2.6.2 SHARE query client

- **Database-free SPARQL queries**
- **Web-based client**
  - SPARQL-assist query composer
- **Command-line client**
  - Java-based
    (executable JAR)
  - platform-agnostic

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2.6.3 SADI Taverna plugin

- **Adding SADI services to a workflow**
  - integration with non-SADI services
- **Configuring SADI services**
  - combining input from multiple services
  - extracting output
- **Discovering SADI services**
  - expanding an existing workflow

Thank you !



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SADI Semantic Web Services

Instructor: Luke McCarthy

http:// sadiframework.org/training/

canarie